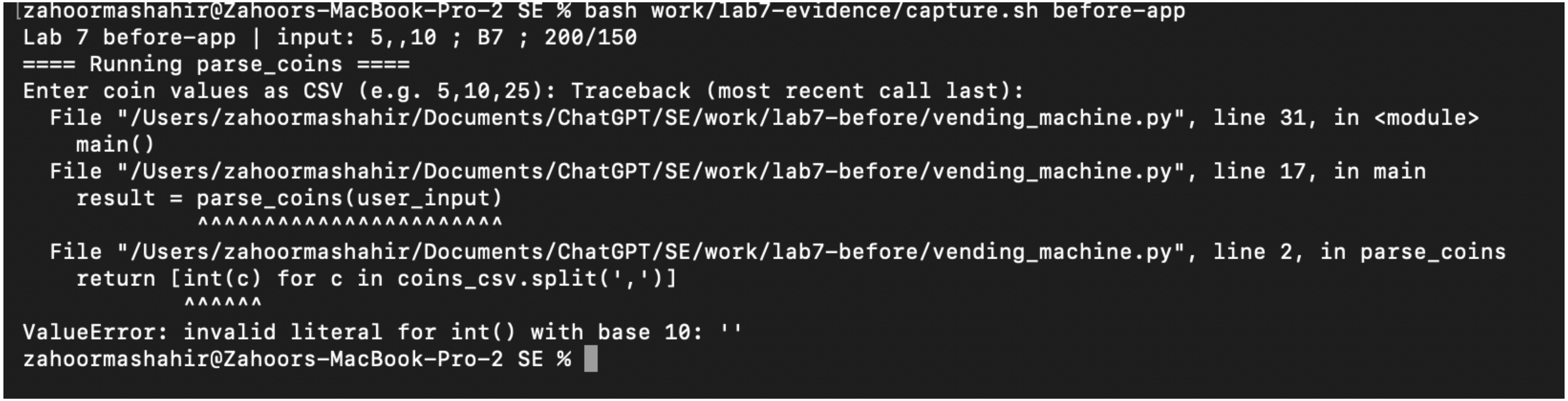
Before the fixes

Vending Machine | PES1UG24CS579

01 Original application

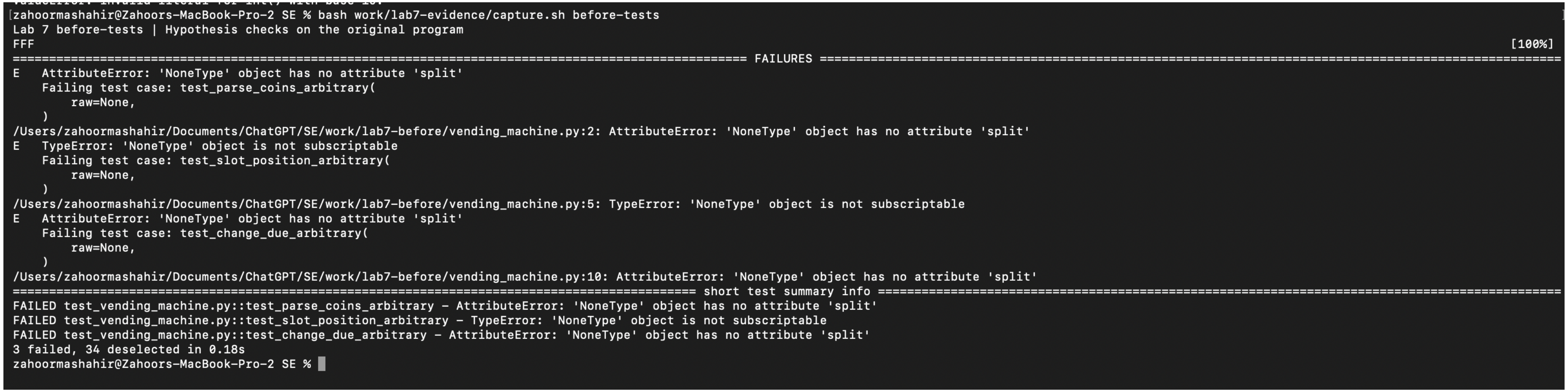
The empty token in 5,,10 raises ValueError and stops the program at the first prompt.



```
[zahoormashahir@Zahoors-MacBook-Pro-2 SE % bash work/lab7-evidence/capture.sh before-app
Lab 7 before-app | input: 5,,10 ; B7 ; 200/150
==== Running parse_coins ====
Enter coin values as CSV (e.g. 5,10,25): Traceback (most recent call last):
  File "/Users/zahoormashahir/Documents/ChatGPT/SE/work/lab7-before/vending_machine.py", line 31, in <module>
    main()
  File "/Users/zahoormashahir/Documents/ChatGPT/SE/work/lab7-before/vending_machine.py", line 17, in main
    result = parse_coins(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/zahoormashahir/Documents/ChatGPT/SE/work/lab7-before/vending_machine.py", line 2, in parse_coins
    return [int(c) for c in coins_csv.split(',')]
            ^^^^^^
ValueError: invalid literal for int() with base 10: ''
zahoormashahir@Zahoors-MacBook-Pro-2 SE % 
```

02 Hypothesis checks on the original code

All three selected parser properties fail. Hypothesis reduces each failing input to None.



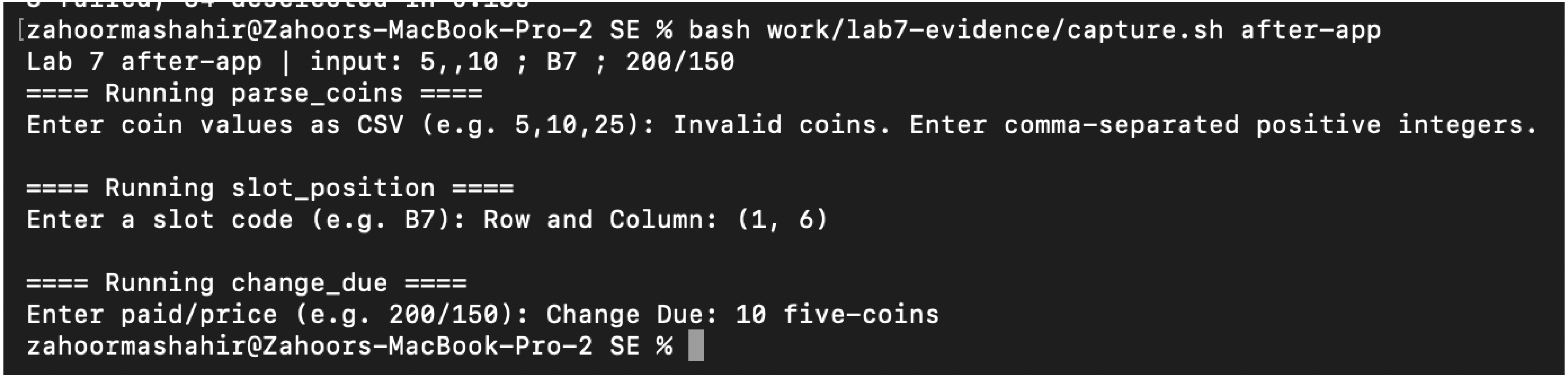
```
[zahoormashahir@Zahoors-MacBook-Pro-2 SE % bash work/lab7-evidence/capture.sh before-tests
Lab 7 before-tests | Hypothesis checks on the original program
FFF                                                                                                                                                                                                           [100%]
===================================================================================================== FAILURES ======================================================================================================
E   AttributeError: 'NoneType' object has no attribute 'split'
    Failing test case: test_parse_coins_arbitrary(
        raw=None,
    )
/Users/zahoormashahir/Documents/ChatGPT/SE/work/lab7-before/vending_machine.py:2: AttributeError: 'NoneType' object has no attribute 'split'
E   TypeError: 'NoneType' object is not subscriptable
    Failing test case: test_slot_position_arbitrary(
        raw=None,
    )
/Users/zahoormashahir/Documents/ChatGPT/SE/work/lab7-before/vending_machine.py:5: TypeError: 'NoneType' object is not subscriptable
E   AttributeError: 'NoneType' object has no attribute 'split'
    Failing test case: test_change_due_arbitrary(
        raw=None,
    )
/Users/zahoormashahir/Documents/ChatGPT/SE/work/lab7-before/vending_machine.py:10: AttributeError: 'NoneType' object has no attribute 'split'
=============================================================================================== short test summary info ===============================================================================================
FAILED test_vending_machine.py::test_parse_coins_arbitrary - AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_vending_machine.py::test_slot_position_arbitrary - TypeError: 'NoneType' object is not subscriptable
FAILED test_vending_machine.py::test_change_due_arbitrary - AttributeError: 'NoneType' object has no attribute 'split'
3 failed, 34 deselected in 0.18s
zahoormashahir@Zahoors-MacBook-Pro-2 SE % 
```

After the fixes

Vending Machine | PES1UG24CS579

03 Fixed application

The same run rejects 5,,10, maps B7 to (1, 6), and returns 10 five-coins for 200/150.



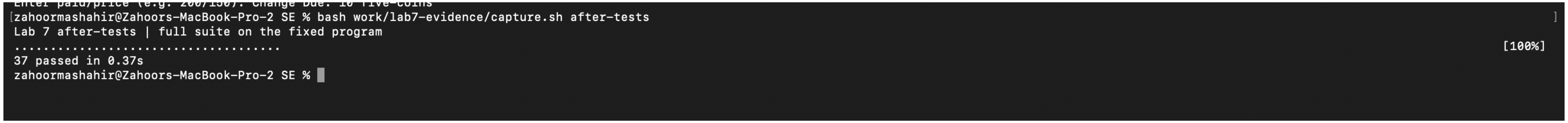
```
[zahoormashahir@Zahoors-MacBook-Pro-2 SE % bash work/lab7-evidence/capture.sh after-app
 Lab 7 after-app | input: 5,,10 ; B7 ; 200/150
 ==== Running parse_coins ====
 Enter coin values as CSV (e.g. 5,10,25): Invalid coins. Enter comma-separated positive integers.

 ==== Running slot_position ====
 Enter a slot code (e.g. B7): Row and Column: (1, 6)

 ==== Running change_due ====
 Enter paid/price (e.g. 200/150): Change Due: 10 five-coins
 zahoormashahir@Zahoors-MacBook-Pro-2 SE %
```

04 Full test suite on the fixed code

All 37 tests pass, including the three properties that failed against the original program.



```
[zahoormashahir@Zahoors-MacBook-Pro-2 SE % bash work/lab7-evidence/capture.sh after-tests
 Lab 7 after-tests | full suite on the fixed program
 .....................................                                                                                          [100%]
 37 passed in 0.37s
 zahoormashahir@Zahoors-MacBook-Pro-2 SE %
```